ADC Documentation

ADC Definition:

The 12-bit ADC is a successive approximation analog-to-digital converter. It has up to 19 multiplexed channels allowing it to measure signals from 16 external sources, two internal sources, and the VBAT channel. The A/D conversion of the channels can be performed in single, continuous, scan or discontinuous mode. The result of the ADC is stored into a left- or right-aligned 16-bit data register. The analog watchdog feature allows the application to detect if the input voltage goes beyond the user-defined, higher or lower thresholds.

1- ADC on/off control : ADC is powered on by setting the ADON bit in the ADC_CR2 register and powered down by clearing the ADON bit

2- ADC clock : it has 2 clock schemes

– Clock for the analog circuitry: ADCCLK ,generated from the APB2 clock

– Clock for the digital interface : equal to the APB2 clock

3- ADC Registers :

3.1 ADC status register (ADC_SR):

Address offset: 0x00 , Reset value: 0x0000 0000

– tells the status of the program

– most important flag: is (EOC) end of conversation flag as it checks if the conversation ended.

3.2 ADC control register 1 (ADC_CR1) :

Address offset: 0x04 , Reset value: 0x0000 0000

– configures ADC behavior

– most important flags : RES selects the resolution of the conversion, SCAN to enable/disable the Scan mode. In Scan mode, the inputs selected through the ADC_SQRx or ADC_JSQRx registers are converted.

3.3 ADC control register 2 (ADC_CR2) :

Address offset: 0x08 , Reset value: 0x0000 0000

– controls how the ADC operates .

– most important bits : {SWSTART} starts conversion and cleared by hardware as soon as the conversion starts,, {CONT } If it is set, conversion takes place continuously until it is cleared ,, { ADON } converter ON / OFF

3.4 ADC sample time register 1 (ADC_SMPR1) & ADC sample time register 2 (ADC_SMPR2) :

1- Address offset: 0x0C , Reset value: 0x0000 0000

2- Address offset: 0x10 , Reset value: 0x0000 0000

– Select the sampling time individually for each channel.

3.5 ADC regular sequence register 1 (ADC_SQR1) & ADC regular sequence register 2 (ADC_SQR2) & ADC regular sequence register 3 (ADC_SQR3) :

1- Address offset: 0x2C , Reset value: 0x0000 0000

2- Address offset: 0x30 , Reset value: 0x0000 0000

3- Address offset: 0x34 , Reset value: 0x0000 0000

– controls the order of the channels used

3.6 ADC regular data register (ADC_DR) :

Address offset: 0x04 , Reset value: 0x0000 0000

– where the final data is stored .

4- ADC1 Channel Base Address :

ADC1 : 0x4001 2000 - 0x4001 23FF

Note : for the complete sequence of each register's bits ,please go back to the ADC register map in the STM32 manual page 241 & 242